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Enhancing surveillance anomaly detection with keyframes and explainable inception model

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ABSTRACT

The proliferation of surveillance systems in the 21st century has become critical in mitigating rising crime rates and preserving life and property. However, conventional surveillance methods struggle with detecting abnormal human activities in complex environments characterized by varying lighting conditions and diverse appearances of individuals. To address these challenges, we propose advanced computer vision techniques, particularly supervised anomaly detection, to enhance intelligent surveillance systems. This research introduces SilentFrm to optimize anomaly detection by analyzing frame differences and setting thresholds based on calculated areas. SilentFrm streamlines the keyframe selection process, enhancing efficiency and reducing redundancy in anomaly detection tasks. Additionally, we present XAI-Inv3, an innovative architecture that integrates interpretability techniques such as Grad-CAM and guided backpropagation into the InceptionV3 model. These methodologies exhibit high accuracy and provide interpretable insights, thereby enhancing situational awareness and security. Rigorous experimentation showcases SilentFrm and XAI-Inv3′s superior detection accuracy and interpretability compared to state-of-the-art methods. By leveraging insights from weak supervision and semi-supervised learning, our approach aims to develop robust anomaly detection models capable of discerning anomalies with precision and efficiency in dynamic surveillance environments. Integrating XAI-Inv3 with the keyframes approach facilitates real-time anomaly detection and ensures scalability and feasibility across large-scale surveillance deployments. The proposed model achieved an average accuracy of 99 % on Hockey Fight, Violent Flow and Real-Life Violence Situation datasets for violence detection and 92 % for violence recognition using UCF-Crime and Shanghai Tech datasets. Proposed contributions underscore the importance of innovative methodologies in advancing the field of surveillance video analysis, ultimately enhancing public safety and crime prevention efforts.

1. Introduction

In the 21st century, the proliferation of surveillance systems has become critical in mitigating rising crime rates, facilitating preserving life and property. However, conventional surveillance methods often struggle to effectively detect abnormal human activities, particularly in complex environments characterized by varying lighting conditions and diverse appearances of individuals. To address these challenges, researchers have turned to intelligent surveillance systems empowered by computer vision technology [1]. These systems leverage advanced algorithms to automatically identify anomalous events in areas such as hospitals, airports, and parks where surveillance cameras are deployed.

The advent of intelligent surveillance systems heralds a new era in crime prevention and public safety. In recent years, the deployment of surveillance cameras in various public and private domains has become ubiquitous [2]. Yet, the mere presence of cameras is insufficient in thwarting criminal activities. Traditional surveillance methods, reliant on manual monitoring or rudimentary algorithms, often falter in accurately identifying suspicious behaviors amidst the deluge of visual data captured by these systems [3]. The primary obstacle to effective surveillance lies in the complexity of real-world environments. Illumination changes, occlusions, and the inherent diversity in human appearances

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pose formidable challenges to existing surveillance technologies [4]. Consequently, there exists an imperative to develop innovative approaches capable of discerning anomalies with precision and efficiency.

Conventional surveillance methods predominantly rely on predefined rules or simple algorithms to detect anomalies. However, such approaches often yield high false-positive rates and are susceptible to environmental variations. For instance, fluctuating lighting conditions can obscure crucial visual cues, leading to erroneous detections or missed anomalies. Moreover, the diversity in human appearances further exacerbates the challenge, as traditional models struggle to generalize across varied demographics and clothing styles. Furthermore, the computational complexity of processing vast volumes of video data in real time presents a significant bottleneck. Traditional surveillance systems are often resource-intensive, requiring substantial computing power and storage capacity to analyze footage continuously [5]. This not only escalates the operational costs but also hampers the scalability of surveillance deployments.

One of the primary challenges plaguing existing surveillance systems is their limited ability to generalize across diverse environmental conditions and human behaviors [6]. Traditional models often struggle to adapt to lighting, weather, and crowd dynamics variations, leading to increased false alarms and missed anomalies. An inherent imbalance between normal and abnormal instances within surveillance datasets poses a significant challenge to anomaly detection algorithms. In many real-world scenarios, anomalous events are rare occurrences, resulting in skewed class distributions that can undermine the performance of machine learning models. The multifaceted nature of anomalies further complicates detection efforts. Anomalies may manifest in various forms, ranging from subtle deviations in behavior to overt acts of violence, making it challenging for surveillance systems to discern between benign and malicious activities accurately. Real-time anomaly detection in surveillance videos demands significant computational resources, posing scalability challenges for existing methodologies. High computational overhead escalates operational costs and limits the feasibility of deploying surveillance systems across large-scale environments [7]. Achieving real-time processing capabilities while maintaining high detection accuracy remains a formidable challenge in surveillance applications. Delays in anomaly detection can compromise the effectiveness of surveillance systems, particularly in time-sensitive scenarios such as law enforcement interventions or emergency response situations [8].

To surmount these challenges, researchers are increasingly turning towards advanced computer vision techniques, particularly supervised anomaly detection, to bolster the capabilities of surveillance systems. By leveraging machine learning algorithms, these approaches can learn intricate patterns from normal and abnormal behaviors, enhancing the system's ability to discern anomalies amidst visual noise. One promising avenue of research involves weak supervision techniques, which necessitate minimal labeled data for model training. These methods hold immense potential in reducing the annotation burden while achieving comparable performance to fully supervised approaches. Semi-supervised techniques, such as trajectory-based analysis and dictionary learning, offer another avenue for enhancing anomaly detection by capturing nuanced motion patterns and visual features [9].

In pursuit of cost-effective solutions for real-time anomaly detection in surveillance videos, this research article advocates for exploring novel methodologies that strike a balance between computational efficiency and detection accuracy. Drawing inspiration from recent advancements, the proposed approach seeks to leverage insights from weak supervision and semi-supervised learning to develop robust anomaly detection models. Moreover, the integrating frameworks holds promise in streamlining surveillance operations by enabling faster and more accurate analysis of video streams. Models such as SilentFrm and XAIViolence exemplify the potential of deep learning in enhancing violence detection and anomaly recognition in large-scale video databases.

In our work to enhance the interpretability and efficacy of existing deep learning architectures for anomaly detection in surveillance videos, we have devised a novel adaptation termed XAI-Inv3 (eXplainable AI based on InceptionV3). Building upon the foundational architecture of InceptionV3, renowned for its prowess in image recognition tasks, XAI-Inv3 represents a tailored approach towards addressing the imperative for explainable and efficient anomaly detection in dynamic surveillance environments.

The modification process entails integrating interpretability techniques, notably gradient-based methods, into the architecture of InceptionV3. By augmenting the network with mechanisms such as Grad-CAM (Gradient-weighted Class Activation Mapping) and guided backpropagation, we empower XAI-Inv3 to elucidate the salient features within surveillance video frames that underpin its anomaly detection decisions. This strategic incorporation enables stakeholders to gain valuable insights into the model's decision-making process, fostering trust and facilitating human interpretation of detected anomalies. Furthermore, the utility of XAI-Inv3 is particularly pronounced when coupled with a keyframes approach in surveillance video analysis. Keyframes, representing selectively chosen frames encapsulating significant events or transitions within a video sequence, serve as pivotal reference points for anomaly detection algorithms. By applying XAI-Inv3 to analyze keyframes extracted from surveillance footage, we leverage its interpretability capabilities to discern anomalous behaviours precisely and clearly.

The synergy between XAI-Inv3 and the keyframes approach offers several notable advantages in surveillance video analysis. Firstly, it streamlines computational resources by focusing interpretability efforts on pivotal frames, thereby facilreal-time anomaly detection without compromising accuracy. Secondly, elucidating salient features within keyframes enables rapid and intuitive understanding of detected anomalies, expediting decision-making processes for security personnel and law enforcement agencies.

The contributions of our work are as follows:

- We introduce SilentFrm, a novel method for selecting keyframes in surveillance videos. SilentFrm streamlines the process by analyzing frame differences and setting a threshold based on calculated areas. This approach optimizes keyframe selection, enhancing efficiency in anomaly detection tasks.
- We have devised XAI-Inv3, an architecture integrating interpretability techniques into the InceptionV3 model. By incorporating methods such as Grad-CAM and guided backpropagation, XAI-Inv3 enhances the interpretability of anomaly detection decisions, fostering transparency and human understanding.
- Our research extensively evaluates the performance of SilentFrm and XAI-Inv3 across various datasets, including UCF Crime and Shanghai Tech. Additionally, we extend our evaluation to non-surveillance datasets like Hockey Fight and real-life violence situations, ensuring a comprehensive assessment of their effectiveness.
- Through rigorous experimentation, we demonstrate the effectiveness of SilentFrm and XAI-Inv3 in detecting anomalies across diverse surveillance scenarios. These methodologies exhibit high accuracy and provide interpretable insights into anomaly detection decisions, enhancing situational awareness and security.
- We benchmark SilentFrm and XAI-Inv3 against state-of-the-art anomaly detection methods. Our results showcase their superiority in terms of detection accuracy and interpretability, highlighting their potential as leading solutions in surveillance video analysis.

2. Literature review

Recent research addresses concerns in animation content by proposing a deep learning-based system for real-time detection, with a focusing blood as a key indicator [10]. Leveraging a modified Faster R-CNN model, the system integrates technical enhancements such as a modified RegNet model for feature extraction, a modulated deformable

Table 1
Summary of methods, datasets, metrics, and findings in violence detection research.

Method	Dataset	Performance metric	Results
Modified Faster R-CNN	Custom Animation Dataset	Accuracy	High accuracy in real-time blood detection
Deep Labeller	WVD, USI	IoU	IoU: 0.80036 (Stage 1), 0.95 (Stage 2)
Crime Monitoring System (CMS)	Custom CCTV Dataset	Accuracy	Weapon Detection: 80 %, Violence Detection: 95 %, Face Recognition: 97 %
Conv3D-based ResNet-3D	UBI-Fight, Hockey, Crowd, Movie-Fights	AUC	AUC > 94 % (Perfect 100 % on Movie-Fights Dataset)
1D-CNN for Urdu Tweets	Annotated Urdu Corpus (4808 tweets)	Accuracy, Macro F1-Score	Accuracy: 89.84 %, Macro F1-Score: 89.80 %
SME, STE, and GTE Modules	VioPeru, Benchmark Datasets	Accuracy	Superior performance on VioPeru dataset
CrimeNet with Vision Transformer (ViT)	Various Violence Detection Datasets	ROC AUC	High accuracy, near-zero false positives
DeepFrame and AreaDiff with KFNet	Various Violence Detection Datasets	Accuracy	Accuracy: 91 %–99.98 %
3D CNN with Attention Mechanism	UCF Crime Dataset	Accuracy	Higher accuracy on UCF Crime dataset compared to existing methods
Nonlinear Auto-Regressive Neural Network	Green Gram Prices Dataset (2010–2020)	RMSE	RMSE (Training/Validation/Testing): 4.34 %, 4.71 %, 3.98 %
Nonlinear Auto-Regressive Neural Network	Energy Commodity Dataset (2010–2021)	RMSE	Errors: 1.78 %–21.76 %
Nonlinear Auto-Regressive Neural Network	Carbon Emission Allowances (2013–2021)	RMSE	Accurate and stable results
Gaussian Process Regression	Yellow Corn Prices (2010–2020)	Relative RMSE	Relative RMSE: 1.245 %
Gaussian Process Regression	Steel Price Index (2010–2021)	Relative RMSE	Relative RMSE: 0.5432 %
Gaussian Process Regression	Pre-Owned Housing Prices (2012–2020)	Relative RMSE, Correlation	Relative Errors: 0.0458 %–0.3035 %, Correlation: Up to 99.97 %

convolutional (MDC) [11] layer for deformable feature maps, and a distributed attention module (DAM) [12] for improved performance [13]. Additionally, multiscale Region of Interest (ROI) Align techniques are employed for enhanced detection across various scenarios, complemented by classification approaches to categorize violence levels. The system's efficacy is underscored by its ability to achieve high accuracy in real-time blood detection. It facilitates content regulation efforts by online platforms and government bodies, thereby ensuring age-appropriate viewing experiences for children.

Manual labelling of violence in image datasets is laborious and error-prone, necessitating an automated solution. Deep Labeller, a two-stage deep learning approach, leverages pre-trained object detection models and synthetic images to generate "weak labels" in the first stage, subsequently refining the model on these labels in the second stage [14]. Results demonstrate high Intersection over Union (IoU) scores, with 0.80036 achieved in stage 1 and 0.95 in stage 2 on the WVD dataset. Moreover, Deep Labeller exhibits strong generalizability, evidenced by a mean IoU of 0.7450 on the USI dataset, underscoring its potential to streamline violence detection labelling processes while addressing ethical concerns associated with manual labelling practices.

Manual monitoring of CCTV footage is hindered by human limitations, prompting the development of the Crime Monitoring System (CMS), which utilizes deep learning and image processing for real-time crime detection. The system integrates deep learning models, including YOLOv5 for weapon detection, MobileNetv2 for violence detection, and a face recognition algorithm [15]. Through transfer learning and training on image datasets, CMS achieves impressive accuracy rates: over 80 % for weapon detection, 95 % for violence detection, and 97 % for face recognition. The system's performance in real-world scenarios underscores its efficacy in generating timely alerts for crime incidents, thereby improving security and safety by automating crime detection and reducing reliance on human monitoring. Overall, this research highlights the potential of deep learning to revolutionize crime detection, enhancing public safety through efficient and accurate surveillance systems.

Accurately detecting violent behavior in real-world CCTV footage poses a significant challenge due to location and camera specifications variations. To address this, a novel approach leveraging deep learning is proposed, focusing on analyzing the spatiotemporal context of videos to capture both spatial and temporal features [16]. This method integrates optical flow and RGB data to represent motion and color information, respectively, utilizing a Conv3D-based ResNet-3D model as the core network. An attention mechanism is also incorporated to prioritize crucial frames during violent events. The proposed model exhibits state-of-the-art performance across various violence detection datasets, including UBI-Fight, Hockey, Crowd, and Movie-Fights. Exceptionally high Area Under the Curve (AUC) scores exceeding 94 % are achieved on all datasets, with a perfect score of 100 % attained on the Movie-Fights dataset.

The proliferation of unmonitored social media data poses a significant concern due to its potential to disseminate hate speech and incite violence. However, existing automatic detection systems have largely overlooked the Urdu language, leaving a gap in addressing violence incitement in Urdu social media posts [17]. To bridge this gap, a novel framework is proposed to identify expressions of violence incitement within Urdu tweets, leveraging semantic analysis, word embeddings, language models, and a 1-Dimensional Convolutional Neural Network (1D-CNN) for text classification. Trained on a newly curated annotated Urdu corpus comprising 4808 tweets sourced from Pakistani Twitter accounts, the 1D-CNN model utilizing word uni-grams achieved exceptional performance. Notably, it attained an accuracy of 89.84 %, a macro f1-score of 89.80 %, and an f1-score of 89.76 % for identifying violence-inciting tweets specifically. Additionally, it achieved an f1-score of 89.84 % for identifying non-violence tweets, outperforming alternative models such as BiLSTM, CBi-LSTM, and pre-trained language models like Urdu-BERT and Urdu-RoBERTa.

Real-time violence recognition in video surveillance is paramount for bolstering public safety, yet prevailing methodologies often sacrifice efficiency for high accuracy, impeding their practical deployment. To address this, a novel model comprising three specialized modules is proposed: the Spatial Motion Extractor (SME) identifies areas of interest with movement, the Short Temporal Extractor (STE) analyzes rapid movements within these regions [18], and the Global Temporal Extractor (GTE) discerns longer-term patterns to enhance accuracy. Successful evaluations on benchmark datasets and validation on a newly introduced real-world dataset, VioPeru, highlight the model's superior performance compared to existing methods, affirming its potential for real-time applications in video surveillance.

Efficiently analyzing vast video data for violence detection, particularly in smart city contexts, poses computational challenges necessitating high accuracy solutions. Traditional methods reliant on manually engineered features have shown limitations in efficiency, prompting a shift towards deep learning models. Specifically, deep sequence learning approaches are gaining prominence for their potential in real-time

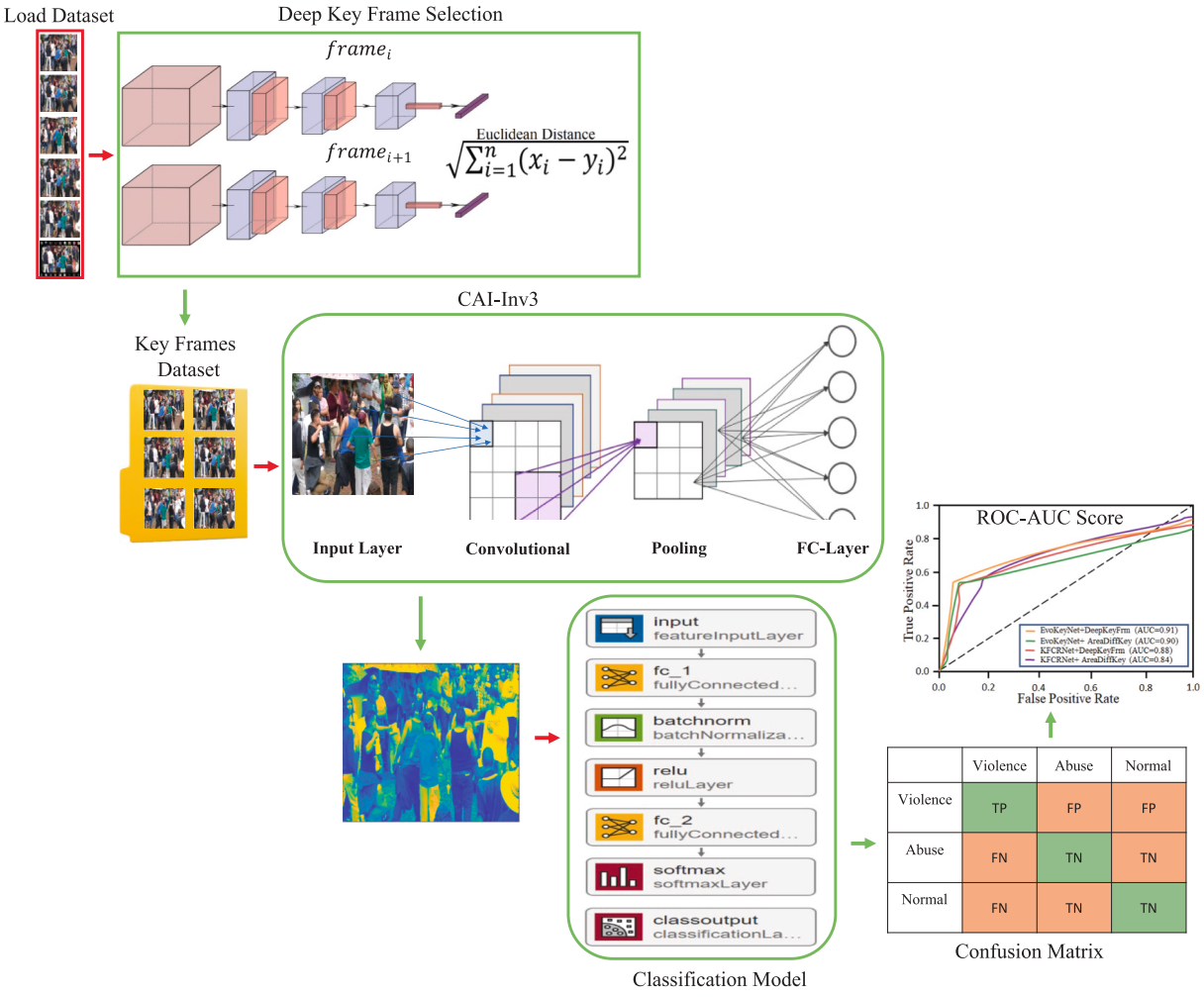


Fig. 1. Proposed explainable AI model for violence detection.

violence detection analysis. Moreover, discussions in this overview extend to localization strategies for pinpointing violence within videos, alongside exploring the trade-offs between simpler machine learning methods and complex deep learning models [19]. Through an analysis of existing datasets, the review sheds light on their strengths and weaknesses, informing future directions in violence detection research. These insights advocate for advancements that balance computational efficiency with enhanced accuracy to tackle the complexities of Big Video Data analysis in smart city environments.

The study uses a nonlinear auto-regressive neural network to forecast green gram prices in the Chinese market (2010–2020). The model achieves high accuracy, outperforming traditional methods with root mean square errors of 4.34 %, 4.71 %, and 3.98 % for training, validation, and testing. The findings provide valuable insights for price forecasting and policy development in agricultural markets. The study forecasts energy commodity prices using nonlinear autoregressive neural networks, achieving high accuracy with errors between 1.78 % and 21.76 %. The results support technical analysis and policy development.

The study uses a nonlinear autoregressive neural network to forecast carbon emission allowance prices in China (2013–2021), achieving accurate and stable results. The findings support technical forecasts and policy analysis for energy cost management and green transition planning (see Table 1).

This study forecasts China's yellow corn prices (2010–2020) using a Gaussian process regression model optimized with cross-validation and Bayesian techniques. The model achieves high accuracy, with a relative root mean square error of 1.245 %, aiding market evaluation and policy-making. This study forecasts northeast China's daily steel price index (2010–2021) using Gaussian process regression with cross-validation and Bayesian optimization, achieving a relative root mean square error of 0.5432 %. The model supports pricing analysis and decision-making for investors and policy makers. This study forecasts pre-owned housing prices in ten major Chinese cities (2012–2020) using Gaussian process regression with Bayesian optimization and cross-validation. The models achieve high accuracy, with relative errors of 0.0458 %–0.3035 % and correlations up to 99.97 %, aiding market analysis and policy research.

Violent Behavior Detection (VioBD) is a critical component of intelligent video surveillance systems, ensuring public safety by identifying aggressive behaviors [20]. This review systematically explores existing VioBD approaches, categorizing them into traditional, end-to-end deep learning, and hybrid frameworks to understand their principles and challenges. Evaluation of these methods on publicly available datasets enables performance comparisons, shedding light on their strengths and limitations. Furthermore, the paper delineates remaining problems in VioBD research. It forecasts future trends, offering valuable insights to guide the development of more effective and robust VioBD systems for enhanced public safety.

CrimeNet, a novel neural network tailored for video violence detection, introduces advancements in architecture and training techniques to achieve high accuracy and minimize false positives [21]. Leveraging Vision Transformer (ViT) and Neural Structured Learning (NSL) with adversarial training, CrimeNet offers a robust framework for violence detection. CrimeNet surpasses previous methodologies in

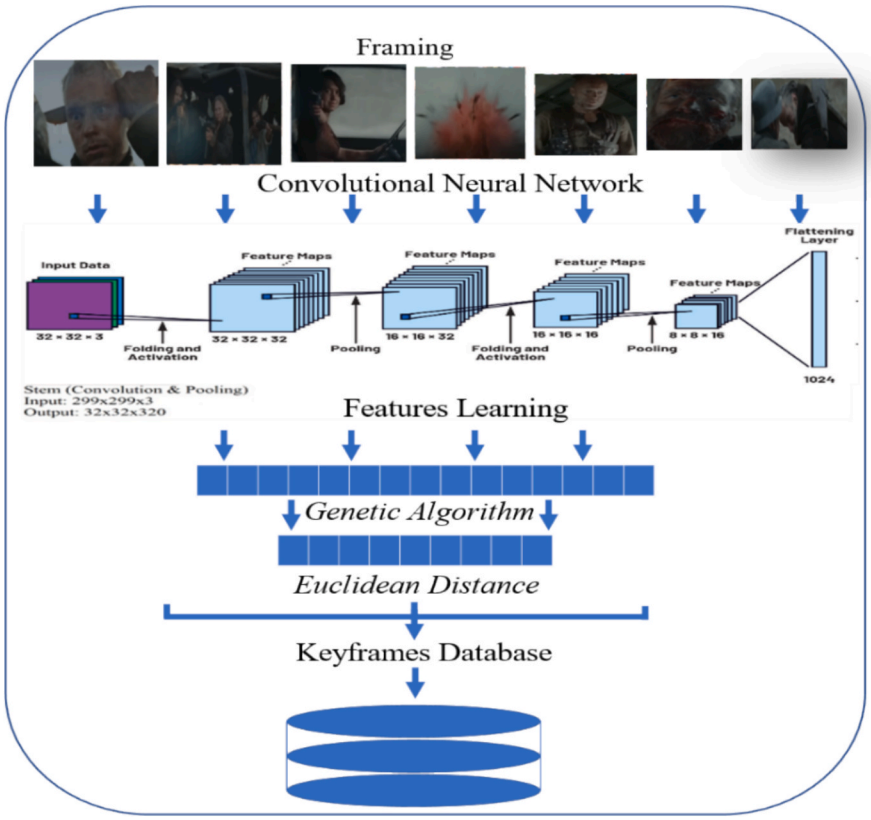


Fig. 2. Proposed keyframes selection framework.

violence detection, notably reducing false positives to near zero while maintaining high accuracy. Evaluated on four challenging violence detection datasets, including binary and multi-class scenarios, CrimeNet exhibits substantial improvements in Receiver Operating Characteristic Area Under the Curve (ROC AUC), a key metric for classifier performance, compared to existing methods. Furthermore, CrimeNet demonstrates robustness in generalizing across diverse datasets, highlighting its efficacy in real-world applications of violence detection in videos.

The proposed system introduces two keyframe-based models, DeepFrame and AreaDiff, demonstrating their efficacy in efficiently selecting keyframes for violence detection. Furthermore, EvoNet employs an evolutionary algorithm to select optimal features, while KFNet combines LSTM, Bi-LSTM, and GRU models for classification, achieving superior accuracy. The proposed system outperforms existing methods, achieving remarkable accuracy ranging from 91 % to 99.98 % across various violence detection datasets. Notably, the system's computational efficiency, facilitated by its focus on keyframes, contributes to its success. Statistical tests including ANOVA and Tukey confirm the significant advancements over previous methodologies, validating the system's efficacy in violence detection within large video databases.

This paper addresses the challenge of automatically detecting suspicious or criminal activity in video surveillance footage, a critical aspect of maintaining public safety and security. The proposed solution introduces an anomaly detection method utilizing an attention mechanism coupled with a 3D Convolutional Neural Network (3D CNN), offering a novel approach to identifying potential crime events in videos. The key steps of the proposed method include background subtraction (BG) to isolate areas of motion, an attention mechanism to prioritize these motion regions, and 3D CNN classification to discern normal activity from anomalies. Remarkably, the system achieves a higher accuracy rate on a public crime event dataset (UCF crime dataset), outperforming existing methods. Additionally, the system's ability to generate alerts for potential crimes enables authorities to take swifter action in response to suspicious activities captured in surveillance footage.

Previous studies have developed advanced systems for real-time violence detection using models like Faster R-CNN, YOLOv5, and Conv3D-based ResNet-3D and frameworks for automating dataset labeling. While achieving high accuracy, these systems face challenges in generalization across diverse environments, balancing efficiency with accuracy, and addressing ethical concerns. This work proposes a scalable, real-time framework that integrates multimodal data, leverages advanced architectures like Vision Transformers, and emphasizes ethical AI practices to enhance practicality and performance.

3. Methodology

The methodology employed in developing and evaluating SilentFrm and XAI-Inv3 for anomaly detection in surveillance videos. Firstly, we describe the implementation of SilentFrm, our novel keyframe selection method, which utilizes frame difference analysis and thresholding techniques to streamline the process efficiently. Subsequently, we delve into the architecture of XAI-Inv3, designed to enhance interpretability in anomaly detection decisions by integrating interpretability techniques such as Grad-CAM and guided backpropagation into the InceptionV3 model. We then outline the comprehensive evaluation framework employed, encompassing diverse datasets including UCF Crime, Shanghai Tech, Hockey Fight, real-life violence situations and violentflow, ensuring a thorough assessment of the methodologies' effectiveness across various surveillance scenarios. Finally, we benchmark SilentFrm and XAI-Inv3 against state-of-the-art anomaly detection methods to demonstrate their superiority in terms of detection accuracy and interpretability, thus affirming their potential as leading solutions in surveillance video analysis. The proposed methodology is exhibited in Fig. 1.

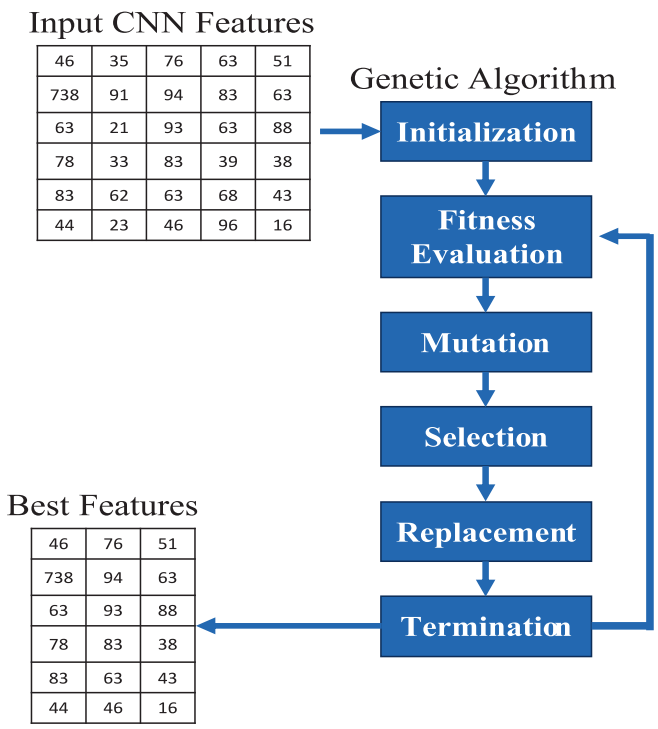


Fig. 3. Proposed GA based best CNN features selection.

3.1. Keyframes selection using GAKeyFrm

This section discusses GAKeyFrm methodology, which combines the InceptionV3 convolutional neural network (CNN) capabilities with a genetic algorithm to optimize feature selection for enhanced anomaly detection performance. Fig. 2 illustrates the methodology for processing consecutive video frames using CNN-based feature extraction. Features are extracted from all video frames using a CNN. A genetic algorithm is then employed to select the optimal CNN features for each frame. Finally, the Euclidean distance between consecutive frames is calculated, and frames with a distance greater than a predefined threshold are selected as keyframes, effectively capturing the most significant frames in the video.

3.1.1. Chromosome representation

Each solution is a binary string indicating selected features:

$$x = [x_1, x_2, \cdots, x_n], x_i \in \{0, 1\}$$

3.1.2. Fitness function

Balances model accuracy with the number of features selected:

$$Fitness = Accuracy(x) - \lambda . \sum_{i=1}^{n} x_i$$

λ: penalty factor (e.g., 0.01)

3.1.3. Mutation operation

Randomly flips bits with a small probability p_m:

$$x_i' = \begin{Bmatrix} 1 - x_i, if random(1) < p_m \\ x_i otherwise \end{Bmatrix}$$

3.1.4. Feature extraction with InceptionV3

GAKeyFrm begins by leveraging the powerful feature extraction capabilities of the InceptionV3 CNN. InceptionV3 is a deep learning architecture trained on a large-scale image dataset, capable of extracting high-level features from input images. Mathematically, given an input image I, InceptionV3 computes a feature vector $F(I)$ representing the image's characteristics through successive convolutional and pooling layers.

Let the input image be denoted as:

$I . \in R^{H \times W \times C}$

H, W and C are the height, width, and number of channels (RGB → C = 3).

$F(I) = InceptionV3(I) = . \varnothing_L(\varnothing_{L-1}(\cdots \varnothing_1(I) \cdots))$

- $\varnothing_l$: transformation at the lth layer (Convolution, ReLU, pooling, etc.)
- L : total number of layers
- Output: Feature vector $F(I) \in R^d$, where d is the dimensionality of the final global feature descriptor (e.g., 2048)

Each convolutional layer applies filters over its input:

$$F^{(l)} = \sigma\left(W^{(l)} * F^{l-1}\right) + b^{(l)}$$

- *: convolution operation
- $W^{(l)}$: learnable filters at layer l
- $b^{(l)}$: biases
- σ: activation function (e.g., ReLU)

3.1.5. Genetic algorithm for feature selection

To optimize feature selection, GAKeyFrm employs a genetic algorithm (GA) that mimics the process of natural selection to iteratively evolve a population of feature subsets. The genetic algorithm operates on a population of candidate feature subsets, with each subset represented as a binary chromosome encoding the presence or absence of features. The genetic algorithm proceeds through the following steps:

1. Initialization
   - Let P represent the population of the candidate feature subsets.
   - Initialize P with N randomly generated chromosomes, where each chromosome $c_i \in P$ is a binary vector of length m (the total number of features).

$$P = \{c_1, c_2, \cdots, c_N\}$$

2. Fitness Evaluation
   - Define a fitness function $f(c)$ that measures the performance of a candidate subset c in the anomaly detection task.
   - The fitness $f(c_i)$ of each candidate subset c_i is evaluated based on its ability to discriminate between normal and anomalous frames.

$$f(c_i) = \text{Performance_Metric}(C_i)$$

3. Selection
   - Select candidate subsets for reproduction using a selection mechanism such as roulette wheel selection, where the probability p_i of selecting subset c_i is proportional to its fitness.

$$p_i = \frac{f(c_i)}{\sum_{j=1}^{N} f(c_j)}$$

4. Crossover
   - Perform crossover on selected parent chromosomes c_a and c_b to produce offsprings $c_{a'}$ and $c_{b'}$.

$$c_{a'} = (c_a[1:k], c_b[k+1:m])$$

$$c_{b'} = (c_b[1:k], c_a[k+1:m])$$

5. Mutation

Introduce random mutations by flipping bits in the chromosomes

Table 2
Genetic algorithm parameters for feature selection in GAKeyFrm.

Parameter	Value
Population Size	100
Crossover Rate	0.8
Mutation Rate	0.1
Number of Generations	50

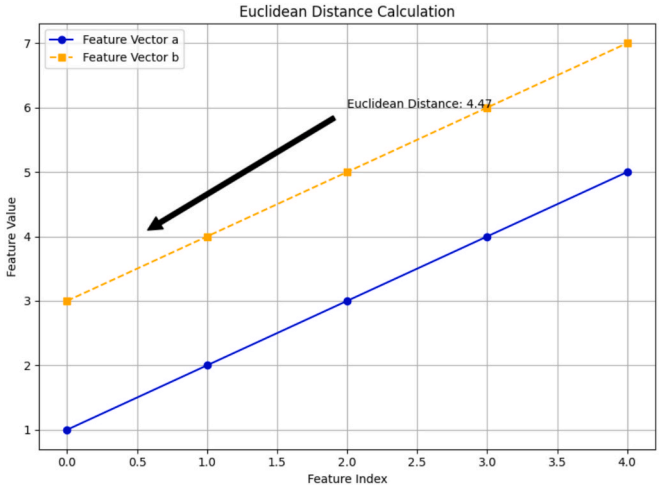


Fig. 4. Euclidean distance calculation process.

with a small probability μ.

$$c_i' = \begin{cases} 1 - c_i[j] with probability \mu \\ c_i[j] with probability (1-\mu) \end{cases} \text{ for each bit } j \text{ in } c_i$$

6. Replacement

Replace the least fit subsets in the population with the newly created offspring.

Sort the population P based on fitness, and replace the least fit $n_{replace}$ individuals.

$$P_{new} = \{top(N - n_{replace}) fit individuals from P\} \cup \{new offspring\}$$

7. Termination

Repeat steps 2–6 for a fixed number of generations G or until a convergence criterion is met (e.g., no significant improvement in fitness). Fig. 3 presents an overview of the GAKeyFrm methodology, depicting the feature extraction process with InceptionV3 and feature selection using a genetic algorithm to enhance anomaly detection in surveillance videos.

The Table 2 presents the genetic algorithm parameters utilized for feature selection in GAKeyFrm. These parameters include the population size, crossover rate, mutation rate, and number of generations. The population size is set to 100, indicating the number of candidate solutions considered in each generation. The crossover rate, representing the probability of crossover occurring between candidate solutions, is set to 0.8. The mutation rate, denoting the probability of a mutation operation being applied to a candidate solution, is set to 0.1. Finally, the number of generations determines the total number of iterations performed during the optimization process, with a value of 50 specified in this case.

$$for g = 1 to G or until convergence$$

3.1.6. *Euclidean distance calculation and threshold selection*

In this subsection, we elucidate the process of applying Euclidean distance to the selected features within GAKeyFrm to quantify their similarity and dissimilarity. The Euclidean distance calculation is a pivotal step in discerning the effectiveness of features for anomaly detection. Mathematically, the Euclidean distance d between two feature vectors a and b in n-dimensional space is computed as:

$$d(a,b) = \sqrt{\sum_{i=1}^{n} (a_i - b_i)^2}$$

Where $a_i and b_i$ represent i-th components of vectors $a and b$ respectively. Following calculating Euclidean distances, GAKeyFrm determines threshold values for distinguishing keyframes from redundant frames. These threshold values are crucial as a decision support system for frame classification. The methodology for selecting threshold values involves calculating distances for a subset of frames, typically the first 1000 frames, and deriving the mean distance μ from these calculations. The threshold distance is then established as a multiple of the mean distance, enabling the classification of frames into keyframes and redundant frames based on their proximity to this threshold. The Fig. 4 demonstrates the computation of Euclidean distance between two sample

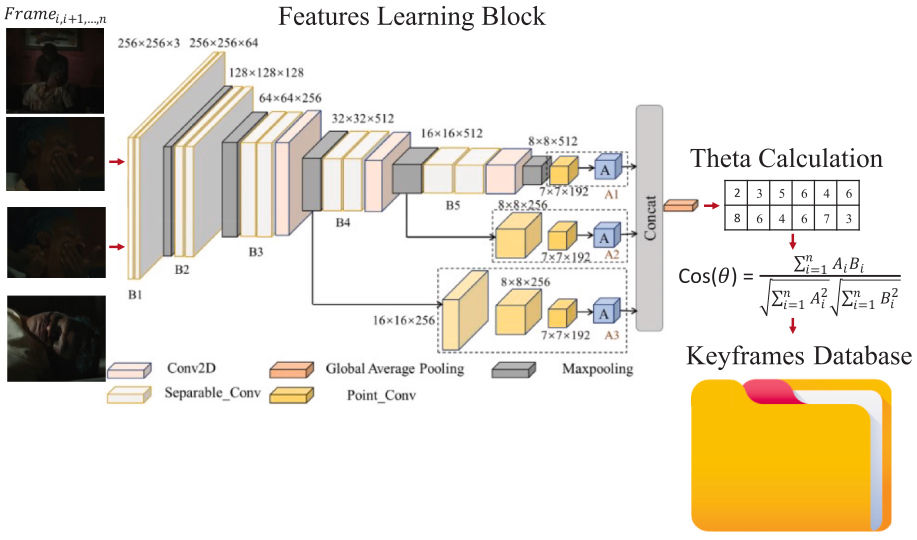


Fig. 5. Decision support system for keyframe classification.

feature vectors (a and b) in GAKeyFrm. The Euclidean distance measures the similarity or dissimilarity between feature vectors, aiding in selecting keyframes for anomaly detection.

This section delves into the decision support system implementation within GAKeyFrm, which is pivotal for classifying frames into keyframes and redundant frames. We elucidate how GAKeyFrm utilizes the calculated threshold distance as a criterion to determine the significance of frames in anomaly detection tasks. Furthermore, we outline the rationale behind using threshold values, emphasizing their role in optimizing the classification process and enhancing the efficiency of anomaly detection in surveillance videos. The flowchart in Fig. 5 demonstrates the process of keyframe classification within the decision support system of GAKeyFrm.

The system utilizes the threshold distance to classify into key and redundant frames, contributing to anomaly detection and surveillance analysis. Decision-making involves evaluating the classification criteria to determine the frames' significance in anomaly detection tasks. This systematic approach enhances the efficiency and effectiveness of anomaly detection in surveillance videos.

3.2. XAI-Inv3 (Explainable InceptionV3)

XAI-Inv3, or Explainable InceptionV3, represents a novel approach to violence detection in surveillance videos by integrating explainable artificial intelligence (XAI) techniques into the InceptionV3 architecture. In traditional deep learning models, the decision-making process often remains opaque, hindering interpretability and trustworthiness. XAI-Inv3 addresses this limitation by providing insight into how the model arrives at its predictions, enhancing transparency and facilitating human understanding.

The architecture of XAI-Inv3 is based on the InceptionV3 convolutional neural network (CNN), a widely used model for image classification tasks. However, XAI-Inv3 incorporates additional layers and techniques to enable interpretability. One key aspect of XAI-Inv3 is its utilization of gradient-based methods, such as Gradient-CAM (Grad-CAM) and guided backpropagation, to visualize the regions of interest in the input image that contribute most to the model's decision.

$$L^{C}_{Grad-CAM} = RELU(\sum_{ij}\frac{\partial Y^{c}}{\partial A_{ij}}w^{c}_{j}A_{ij})$$

In Eq. (1), $L^{C}_{Grad-CAM}$ represents the Grad-CAM for class c,Y^{c} is the predicted class score, A_{ij} denotes the activations in the last convolutional layer, and w^{c}_{j} signifies the weights of the output layer for class c.

3.3. Architecture and components of XAI-Inv3

XAI-Inv3, or Explainable InceptionV3, is a convolutional neural network (CNN) architecture designed for violence detection in surveillance videos. It builds upon the foundation of the InceptionV3 model while incorporating additional components and techniques to enhance interpretability and transparency. The Explainable AI InceptionV3 (XAI-Inv3) model builds on the InceptionV3 architecture by integrating explainability components that allow for deeper insights into the model's decision-making processes. Below is a detailed overview of the architecture and its mathematical underpinnings.

3.3.1. Base InceptionV3 Architecture

InceptionV3 is a deep convolutional neural network (CNN) designed to efficiently handle high-dimensional inputs, leveraging various convolutional and pooling layers to extract features at multiple scales. The architecture comprises several key components:

Input Layer: Accepts images typically sized 299x299x3.
Stem: Initial convolutional layers that reduce dimensionality and extract basic features.
Inception Modules: Series of modules that apply convolutions and pooling operations in parallel, capturing a wide range of features.
Inception module components:
1×1 convolutions
3×3 convolutions
5×5 convolutions
Max pooling and average pooling layers.

$$h_{i,j,k} = \sigma\left(\sum_{m,n}\sum_{d} W_{m,n,d,k}.x_{i+m,j+n,d} + b_{k}\right)$$

Where:

$h_{i,j,k}$ is the output feature map at position (i,j) for the $k-th$ filter.
$W_{m,n,d,k}$ is the weight of the filter at position (m,n,d) for the $k-th$ output feature map.
$x_{i,j,d} =$ is the input at position (i,j) for the $d-th$ channel.
b_{k} is the bias term for the $k-th$ filter.
$\sigma(.)$ is an activation function, typically ReLU.

XAI-Inv3 inherits the basic architecture of the InceptionV3 model, which consists of multiple convolutional and pooling layers followed by fully connected layers. These layers are responsible for extracting hierarchical features from input images. Each Inception module processes the input through multiple paths, performing different types of convolutions or pooling operations and concatenates the outputs.

$$Output = Concat(Branch_{1}(x), Branch_{2}(x), \cdots, Branch_{N}(x))$$

For example, if a module includes 1x1 and 3x3 convolutions and max pooling, their respective operations are:

$$Branch_{1}(x) = \sigma(W_{1x1}{}^{*}x + b_{1x1})$$

$$Branch_{2}(x) = \sigma(W_{3x3}{}^{*}x + b_{3x3})$$

$$Branch_{3}(x) = MaxPool(x, k = 3)$$

Auxiliary classifiers are added at intermediate layers to aid in training:

$$Loss_{aux} = \frac{1}{N_{aux}}\sum_{i=1}^{N_{aux}} CrossEntropy(Softmax\left(W^{i}_{aux}.h^{i}_{aux}.b^{i}_{aux}\right), y)$$

Where h^{i}_{aux} is the intermediate feature map, W^{i}_{aux} and b^{i}_{aux} are weights and biases for the auxiliary classifier, and y is the true label. The integration XAI-Inv3 architecture combines inceptionv3 with explainability components and is trained using a composite loss function:

$$Loss_{total} = Loss_{main} + \lambda_{aux}Loss_{aux} + \lambda_{XAI}Loss_{XAI}$$

3.3.2. Gradient-based methods

XAI-Inv3 integrates gradient-based methods to visualize the regions of interest in the input image that contribute most to the model's predictions. Two prominent techniques used are:

Gradient-CAM or Grad-CAM (Gradient-Weighted Class Activation Mapping) method computes the gradient of the predicted class score with respect to the activations of the last convolutional layer. It highlights the regions of the input image that are crucial for making the prediction.

Grad-CAM highlights important regions in the input image by computing the gradients of the target class concerning the feature map:

$$\alpha^{c}_{k} = \frac{1}{Z}\sum_{i}\sum_{j}\frac{\partial y^{c}}{\partial A^{k}_{i,j}}$$

Where α^{c}_{k} are the weights, y^{c} is the score for class c, and $A^{k}_{i,j}$ are the activations for the $k-th$ feature map. The Gead-CAM heatmap $L^{c}_{Grad-CAM} =$

Table 3
Keyframe selection statistics for different datasets using DeepFrame and AreaDiff techniques.

Datasets_name	Total_key_frames	Key_frame_selector	Total_key_frames
Hockey Fight	Violent. = 28754	DeepFrame	13,984
		AreaDiff	8473
	Non-Violent = 29567	DeepFrame	11,239
		AreaDiff	6882
Violent Flow	Violent = 16542	DeepFrame	7543
		AreaDiff	4987
	Non-Violent = 10234	DeepFrame	3890
		AreaDiff	2874
Real Life Violence Situation	Violent = 41020	DeepFrame	18,239
		AreaDiff	10,274
	Non-Violent = 37890	DeepFrame	18,902
		AreaDiff	11,463
UCF-Crime	Violent = 129835	DeepFrame	76,893
		AreaDiff	69,147
	Non-Violent = 2387657	DeepFrame	683,421
		AreaDiff	562,389
ShanghaiTech	Violent = 87643	DeepFrame	38,765
		AreaDiff	36,291
	Non-Violent = 201554	DeepFrame	81,232
		AreaDiff	72,918

$ReLU(\sum_k \alpha_k^c A^k)$

3.3.3. Guided backpropagation

In this technique, the gradients of the predicted class score with respect to the input image are computed while preserving only positive gradients. This helps highlight the positive features contributing to the prediction.

1. Interpretability Layers

XAI-Inv3 incorporates additional interpretability layers to facilitate human understanding of the model's decision-making process. These layers enable the visualization of salient features and provide insights into why certain predictions are made.

2. Feature Maps

XAI-Inv3 generates feature maps at different network layers, capturing abstract representations of the input image. These feature maps are instrumental in understanding the hierarchical structure of the learned features.

3. Output Layer

The final output layer of XAI-Inv3 produces predictions regarding the presence of violence in the input video frame. The salient features identified through gradient-based methods and interpretability layers influence the model's decisions.

4. Experimental results

This section presents the experimental results of our proposed methodologies, SilentFrm and XAI-Inv3, for anomaly detection in surveillance videos. SilentFrm efficiently selects keyframes from surveillance videos using frame difference analysis and thresholding techniques. This method reduces redundancy and optimises dataset size for improved training and validation efficiency. Table 3 shows the dataset names, total frames per class, and the amount of keyframes chosen from each class.

XAI-Inv3 integrates interpretability techniques into the InceptionV3 model to enhance transparency and understanding in anomaly detection decisions. Techniques such as Grad-CAM and guided backpropagation facilitate visualization of decision-making processes. GAKeyFrm combines InceptionV3 with a genetic algorithm to optimize feature selection, enhancing anomaly detection performance through effective feature subset selection. Table 4 contains information regarding the keyframe selection method, the quantity of keyframes chosen, and how the frames were split for cross-validation during both model training and performance assessment.

Robust measures, such as Receiver Operating Characteristic (ROC)

Table 4
Summary of datasets, keyframe counts, and training-validation partitioning.

Datasets_name	Key_frame_selector	Total_key_frames	Frame_Training	Frame_Validation
Hockey Fight	DeepFrame	Violent = 13984	9789	4195
		Non-Violent = 11239	7867	3372
	AreaDiff	Violent = 8473	5931	2542
		Non-Violent = 6882	4817	2065
Violent Flow	DeepFrame	Violent = 7543	5280	2262
		Non-Violent = 3890	2723	1168
	AreaDiff	Violent = 4987	3410	1496
		Non-Violent = 2874	2012	862
Real Life Violence Situation	DeepFrame	Violent = 18239	12,767	5472
		Non-Violent = 18902	13,231	5671
	AreaDiff	Violent = 10274	7192	3082
		Non-Violent = 11463	8024	3439
UCF-Crime	DeepFrame	Violent = 76893	53,825	23,068
		Non-Violent = 683421	478,394	205,056
	AreaDiff	Violent = 69147	48,403	20,744
		Non-Violent = 562389	393,672	168,716
ShanghaiTech	DeepFrame	Violent = 38765	27,135	11,629
		Non-Violent = 81232	56,862	24,370
	AreaDiff	Violent = 36291	25,404	10,887
		Non-Violent = 72918	51,043	21,875

Table 5
Hyperparameters for training SilentFrm and XAI-Inv3 models.

Model	Hyper_parameter	Value
SILENTFRM	Total_Epochs	50
	Total_Batch_size	64
	Learing_Rate	0.001
	Drop_out	0.5
	Optimization_Method	Adam
	No of parameters	0.072 million
XAI-INV3	Total_Epochs	50
	Total_Batch_size	32
	Learing_Rate	0.0001
	Drop_out	0.4
	Optimization_Method	Adam
	No of parameters	N/A

Table 6
Model efficiency and performance are compared using the UCF-CRIME dataset.

Model	Training_Time	Accuracy (%)
EvoNet	739	80.16 %
EvoNet-DeepFrame	473	83.81s%
EvoNet-AreaDiff	425	86.37 %
EvoNet-DeepFrame + Evolutionary	280	91.48 %
EvoNet-AreaDiff + Evolutionary	243	90.18 %
KFNet	1068	72.56 %
KFNet-DeepFrame	662	74.25 %
KFNet-AreaDiff	587	76.34 %
KFNet-DeepFrame + Evolutionary	392	89.30 %
KFNet-AreaDiff + Evolutionary	326	86.56 %

curves and statistical tests like Analysis of Variance (ANOVA) and P-test, were used to assess the performance of SilentFrm and XAI-Inv3. These assessments offer valuable perspectives on how well our techniques identify irregularities in security footage.

4.1. Training details

For the training of SilentFrm and XAI-Inv3, we utilized keyframes selected via GAKeyFrm and features extracted using InceptionV3. SilentFrm was trained to optimize keyframe selection, while XAI-Inv3 focused on interpretability in anomaly detection. Both models were evaluated on various surveillance video datasets to ensure robustness and effectiveness.

4.1.1. Training XAI-Inv3

XAI-Inv3 was trained on the same datasets, incorporating explainability techniques to provide interpretability of the result. Table 5 provides a detailed overview of the hyperparameters for both models, aiming to maintain consistency and clarity in our methodology presentation.

4.2. Non_Surveillance dataset

Three well-known benchmark datasets unrelated to surveillance—the Hockey Fight dataset [22], the Violent Flow dataset [23], and the Real-Life Violence Situation (RLVS) dataset [24]—were used to assess our suggested frameworks for violence detection. These datasets were selected to provide diverse and realistic scenarios for testing the effectiveness of our methods outside traditional surveillance settings. The Hockey Fight dataset comprises 1000 video clips capturing violent incidents during National Hockey League matches, with balanced labels distinguishing between “fight” and “no fight” events. The Violent Flow dataset includes 276 video clips featuring violent and non-violent content, captured using various devices such as static cameras and mobile phones, with videos ranging from 25 to 200 frames at a resolution of 320 × 240 pixels. Lastly, the RLVS dataset consists of 2000 videos collected from diverse sources, depicting various violent and non-violent situations across different demographics, ensuring comprehensive representation of real-world scenarios [24]. These datasets collectively enable a thorough evaluation of our frameworks’ performance and generalizability in detecting violence across varied non-surveillance environments.

Table 7
Analysis of the suggested violence detection models trained on the hockey fight dataset.

Model Name	True Positives	False Positives	False Negatives	True Negatives	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	G-Mean	MCC
	Hockey Fight Dataset									
EvoNet + DeepFrame	**54.12**	**0.52**	**0.21**	**45.15**	**0.9929**	**0.9904**	**0.9961**	**0.9933**	**0.9924**	**0.9853**
EvoNet + DeepFrame	53.7	0.61	0.34	45.35	0.9905	0.9887	0.9937	0.9912	0.9902	0.9808
KFNet + DeepFrame	52.95	0.43	0.28	46.34	0.9927	0.9919	0.994	0.993	0.9927	0.9857
KFNet + DeepFrame	53.45	0.58	0.19	45.78	0.9923	0.9892	0.9964	0.9928	0.9919	0.9845
	Violent Flow									
EvoNet + DeepFrame	54.05	0.49	0.25	45.21	0.9926	0.991016	0.995396	0.993201	0.992332	0.9850
EvoNet + AreaDiff	52.8	0.55	0.32	46.33	0.9913	0.989691	0.993976	0.991829	0.991118	0.9825
KFNet + DeepFrame	**54.2**	**0.62**	**0.2**	**44.98**	**0.9918**	**0.98869**	**0.996324**	**0.992492**	**0.991351**	**0.9834**
KFNet + AreaDiff	53.9	0.5	0.28	45.32	0.9922	0.990809	0.994832	0.992816	0.991956	0.9842
	Real Life Violence Situation									
EvoNet + DeepFrame	54.15	0.47	0.23	45.15	0.993	0.991395	0.995771	0.993578	0.9927	0.9858
EvoNet + AreaDiff	54.3	0.6	0.18	44.92	0.9922	0.989071	0.996696	0.992869	0.9917	0.9842
KFNet + DeepFrame	53.85	0.53	0.27	45.35	0.992	0.990254	0.995011	0.992627	0.9917	0.9838
KFNet + AreaDiff	**53.5**	**0.45**	**0.35**	**45.7**	**0.992**	**0.991659**	**0.9935**	**0.992579**	**0.9918**	**0.9839**

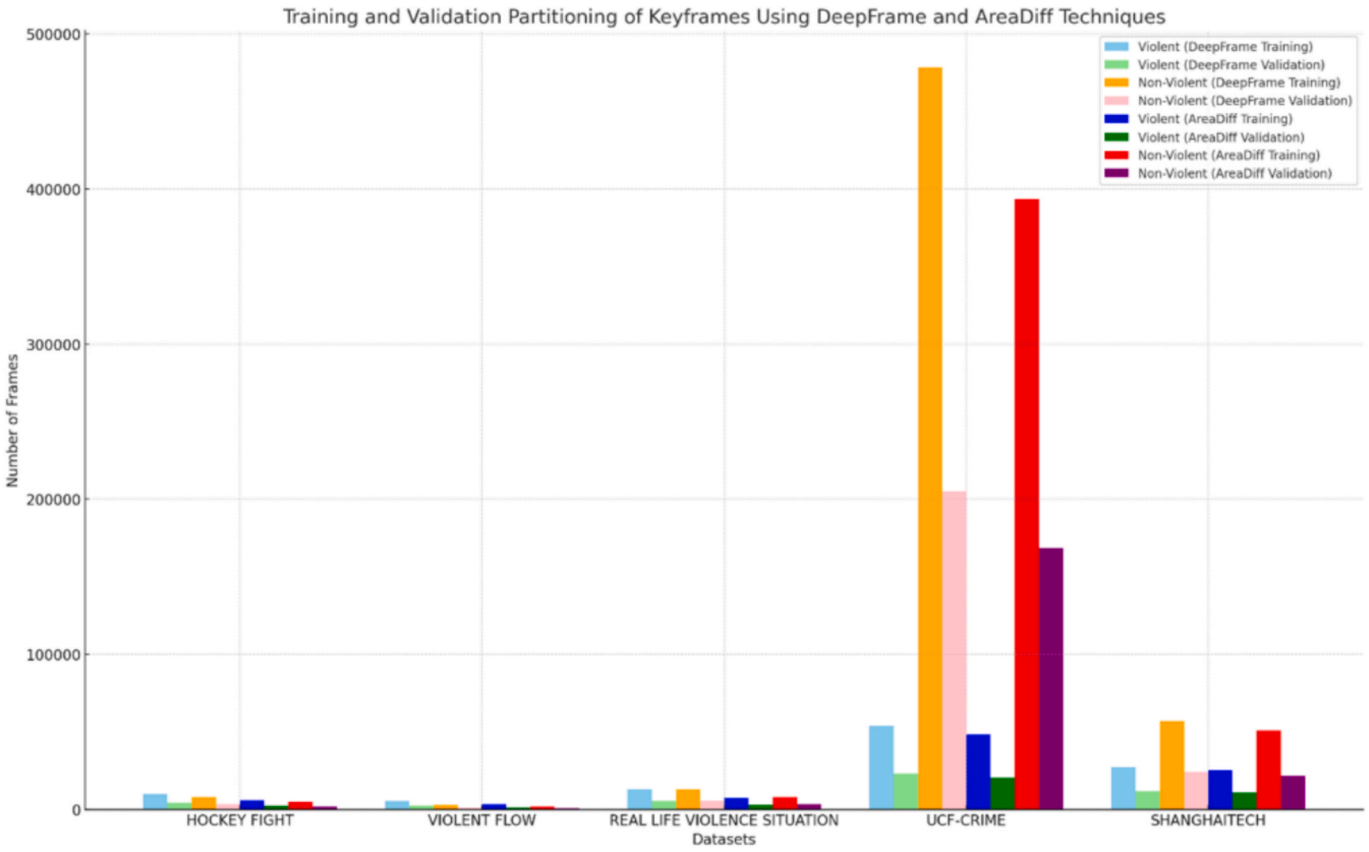


Fig. 6. Training and validation partitioning of keyframes using DeepFrame and AreaDiff techniques.

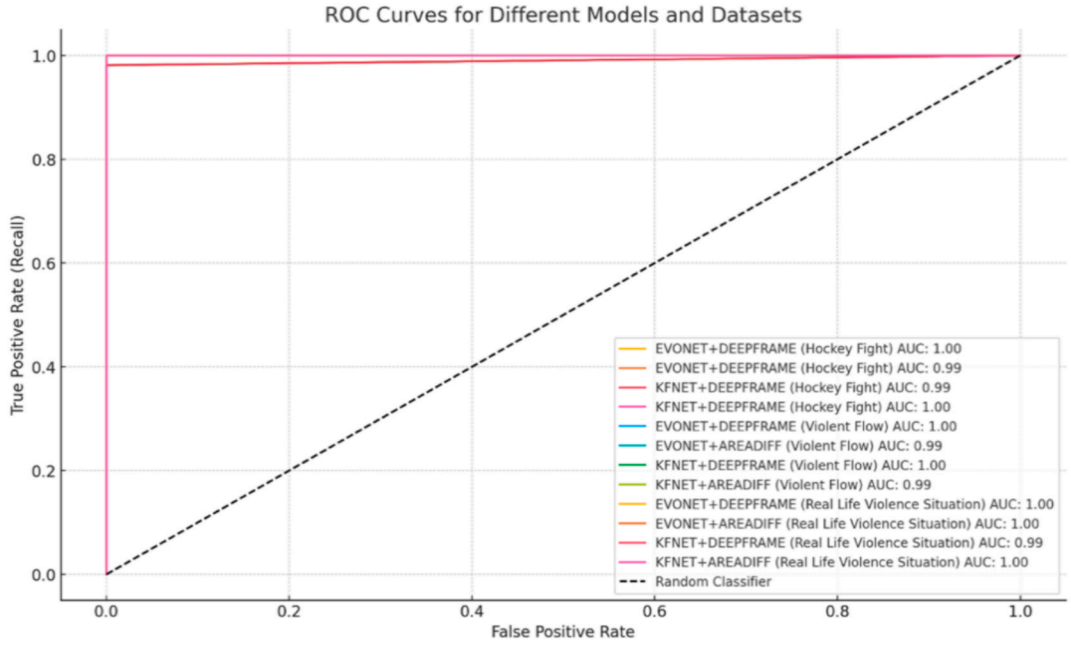


Fig. 7. Comparative ROC analysis of keyframe selection techniques for anomaly detection in surveillance videos.

Table 6 presents a comparative overview of the experimental results obtained from the UCF-Crime Dataset, highlighting the variations in training times and accuracy percentages between different model configurations. These observations improve our comprehension of the many scenarios in which the UCF-Crime Dataset models operate.

4.2.1. *Non_Surveillance datasets results*

Three non-surveillance benchmark datasets—Hockey Fight, Violent Flow, and Real Life Violence Situation (RLVS)—were used to assess the effectiveness of our suggested models for violence detection. Table 7 lists all of the performance indicators in detail for each model, including Matthews Correlation Coefficient (MCC), Accuracy, Precision, Recall, F-

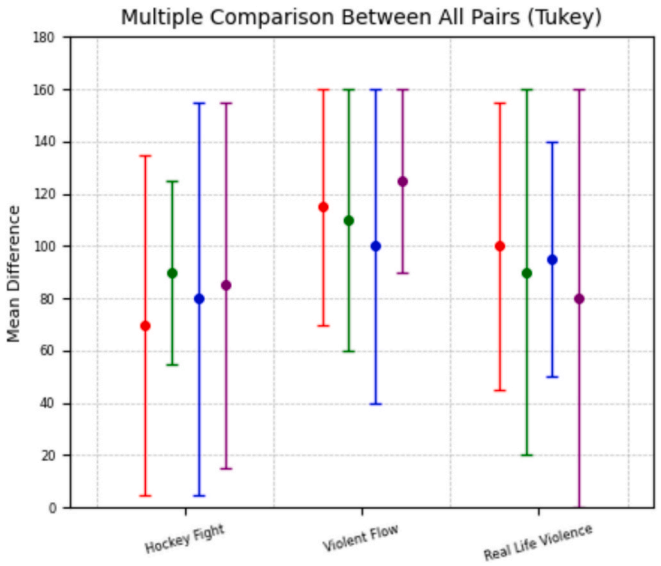


Fig. 8. Multiple comparison of means using Tukey HSD Test.

Measure, G-Mean, and True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN). These datasets, which comprise films classified into violence and non-violence classes, were used to train and evaluate the models EvoNet + DeepFrame, EvoNet + AreaDiff, KFNet + DeepFrame, and KFNet + AreaDiff. All models exhibit great accuracy, with values ranging from 97 % to 99 %, indicating their effectiveness in detecting violent content in video clips. Furthermore, the robust connection between anticipated and actual classifications, with MCC values regularly above 95, highlights the dependability of our algorithms in non-surveillance scenarios.

A thorough comparison of EvoNet and KFNet's performance with the DeepFrame and AreaDiff keyframe selection techniques is shown in Fig. 6. On three particular datasets—Hockey Fight, Violent Flow, and RLVS—the paper examines the mean differences across all pairwise combinations of these designs and methodologies. The Tukey method, a rigorous statistical analytic technique, guarantees comprehensive evaluation. While the x-axis shows mean comparative values for overall performance levels, the y-axis shows mean differences, representing performance variances among combinations. The bars in each dataset show the mean differences for the combinations of EvoNet and DeepFrame, EvoNet and AreaDiff, and KFNet and DeepFrame. Fig. 7 presents a comparative ROC analysis of keyframe selection techniques for anomaly detection in surveillance videos. The graph compares the performance of various models, such as EVONET and KFNET, when combined with DEEPFRAME and AREADIFF methods across multiple datasets, including Hockey Fight, Violent Flow, and real-life violence situations. The ROC curves demonstrate that all models perform exceptionally well, achieving near-perfect AUC scores, indicating the effectiveness of the proposed methods in detecting anomalies in different surveillance scenarios.

In this section, we present the ROC curves and AUC values for various models employed in anomaly detection within surveillance videos. The models evaluated include EVONET + DEEPFRAME, EVONET + AREADIFF, KFNET + DEEPFRAME, and KFNET + AREADIFF, across datasets such as Hockey Fight, Violent Flow, and Real-Life Violence Situation.

The ROC curves depict the trade-off between true positive rate (TPR) and false positive rate (FPR), with AUC values indicating overall effectiveness. Higher AUC values suggest superior performance, reflecting the model's ability to correctly identify anomalies while minimizing false alarms. Models with curves closer to the top left corner exhibit better performance.

Our analysis shows that integrating DeepFrame and AreaDiff techniques with EVONET and KFNET models significantly enhances anomaly detection. These models consistently achieve high AUC values, demonstrating robustness and reliability. This highlights the importance of selecting appropriate keyframe extraction methods and model architectures for optimizing the detection of violent and non-violent

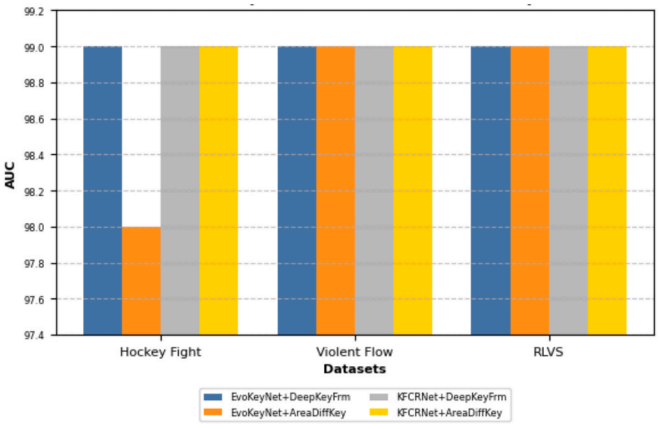


Fig. 9. AUC scores are used to evaluate the performance of the classification model on the hockey fight, violent flow, and RLVS datasets.

Table 8
Tukey_HSD test for multiple comparison of means (FWER = 0.05).

Category_1	Category_2	Mean Difference	P-Value Adjusted	Lower Bound	Upper Bound	Reject
Hockey_fight_Dataset						
EVONET + DEEPFRAME	KFNET + DEEPFRAME	0.22	0.12	−0.11	0.55	FALSE
EVONET + DEEPFRAME	EVONET + AREADIFF	−0.17	0.09	−0.44	0.10	FALSE
KFNET + DEEPFRAME	EVONET + AREADIFF	−0.39	0.03	−0.66	−0.12	TRUE
KFNET + DEEPFRAME	KFNET + AREADIFF	−0.5	0.01	−0.77	−0.23	TRUE
Violent_Flow_Dataset						
EVONET + DEEPFRAME	KFNET + DEEPFRAME	−0.15	0.22	−0.42	0.12	FALSE
EVONET + DEEPFRAME	EVONET + AREADIFF	0.22	0.12	−0.11	0.55	FALSE
KFNET + DEEPFRAME	EVONET + AREADIFF	0.45	0.01	0.12	0.78	TRUE
KFNET + DEEPFRAME	KFNET + AREADIFF	0.35	0.05	0.02	0.68	FALSE
RLVS_Dataset						
EVONET + DEEPFRAME	KFNET + DEEPFRAME	0.28	0.08	−0.05	0.61	FALSE
EVONET + DEEPFRAME	EVONET + AREADIFF	0.13	0.32	−0.20	0.46	FALSE
KFNET + DEEPFRAME	EVONET + AREADIFF	0.20	0.18	−0.13	0.53	FALSE
KFNET + DEEPFRAME	KFNET + AREADIFF+	0.15	0.27	−0.18	0.48	FALSE

Table 9
AUC performance of proposed models on hockey fight, violent flow, and real-life violence situation (RLVS) datasets.

Model	AUC (%)
hockey_fight	
EvoNet-DeepFrame	99 %
EvoNet-AreaDiff	99 %
KFNet-DeepFrame	99 %
KFNet-AreaDiff	99 %
VIOLENT_FLOW	
EvoNet-DeepFrame	99 %
EvoNet-AreaDiff	98 %
KFNet-DeepFrame	99 %
KFNet-AreaDiff	99 %
RLVS	
EvoNet-DeepFrame	99 %
EvoNet-AreaDiff	99 %
KFNet-DeepFrame	99 %
KFNet-AreaDiff	99 %

events.

The Fig. 8 illustrates the mean differences and confidence intervals for multiple comparisons between datasets, including Hockey Fight, Violent Flow, and Real-Life Violence. The Tukey HSD test assessed significant differences between the groups, with a family-wise error rate (FWER) of 0.05. The error bars indicate the variability in mean differences across the datasets, helping to visualize potential significant group differences in anomaly detection performance.

The results of the model comparison for the suggested deep learning architectures using ANOVA and Tukey's multiple comparison tests are shown in Table 8. ANOVA evaluates the general performance of the model over all datasets. As keyframe selection techniques, EvoNet and KFNet appear to perform on par with DeepFrame or AreaDiff, according to the non-significant ANOVA and Tukey results. Given the study's parameters and datasets, this implies that the keyframe selection approach may not significantly impact model performance. The bar chart in Fig. 9 compares the AUC performance of four keyframe selection methods—EvoKeyNet + DeepKeyFrm, EvoKeyNet + AreaDiffKey, KFCRNet + DeepKeyFrm, and KFCRNet + AreaDiffKey—across three datasets: Hockey Fight, Violent Flow, and Real-Life Violence Situations (RLVS). The results demonstrate high AUC values for all methods, indicating their effectiveness in anomaly detection, with slight variations across the datasets and keyframe selection techniques.

Table 9 displays an AUC-metric comparison of the violence detection approach proposed in this study. The analysis of AUC values demonstrates how well the model can identify violent video scenes. Moreover, keyframes and evolutionary search-based techniques greatly enhance the model's detection performance. Three subplots comprised of bar charts illustrate the performance of several model combinations trained on different datasets (Hockey Fight, Violent Flow, and RLVS) in Fig. 7. Subplot (a) displays the performance of four model combinations: KFNet-DeepFrame, KFNet-AreaDiff, EvoNet-DeepFrame, and EvoNet-AreaDiff on the Hockey Fight dataset. Their high AUC values, around 0.99, demonstrate these models' exceptional ability to distinguish between positive and negative cases within the Hockey Fight dataset. The performance of the same model combinations on the Violent Flow dataset is shown in Subplot (b), where their similarly high AUC values near 0.99 demonstrate their excellent discriminatory power. The subplot (c) bar chart illustrates how well these models performed on the RLVS dataset and demonstrates high AUC values of about 0.99. Their consistent performance across datasets shows the models' strong discriminatory abilities in various settings.

4.3. Surveillance dataset results

The UCF_Crime Dataset is widely used to test and compare anomaly detection and video-based action recognition algorithms. It has thirteen distinct classes that stand for different types of criminal activity as well as typical behaviors seen in everyday life. The dataset provides a balanced framework for identifying typical activities by combining non-criminal behaviors with criminal actions like gunshot, assault, robbery, vandalism, and burglary. In the field of computer vision, the ShanghaiTech. Dataset is commonly used as a benchmark for crowd counting and density estimation tasks. In particular, it is. Important for practitioners and academics in areas including crowd analysis, machine learning, and computer vision. The collection includes photos taken in various Shanghai, China public areas, including squares, streets, and public gathering places.

Table 10 thoroughly analyses deep learning models used for violence detection on the ShanghaiTech and UCF-CRIME datasets. Using metrics like True Positive (TP), False Positive (FP), False Negative (FN), True Negative (TN), Accuracy, Precision, Recall, F1-score, G-Mean, and Matthews Correlation Coefficient (MCC), the models evaluated are EvoNet-DeepFrame, EvoNet-AreaDiff, KFNet-DeepFrame, and KFNet-AreaDiff. EvoNet-DeepFrame obtains 90.17 % accuracy, 92.63 % precision, and 89.72 % recall in the UCF-CRIME dataset, while KFNet-DeepFrame displays 87.03 % accuracy, 91.27 % precision, and 91.27 % recall. Comparably, EvoNet-DeepFrame achieves 86.43 % accuracy, 91.64 % precision, and 83.76 % recall on the ShanghaiTech dataset, whereas KFNet-DeepFrame achieves 92.68 % accuracy, 94.29 % precision, and 92.62 % recall..These results underscore the models' effectiveness in distinguishing violent activities across different surveillance scenarios, demonstrating their robust performance in video-based violence detection tasks.

Table 10
Performance metrics of proposed models on UCF-Crime and ShanghaiTech datasets.

Model Name	True Positives	False Positives	False Negatives	True Negatives	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	G-Mean	MCC
	UCF_Crime Dataset									
EvoNet-DeepFrame	50.66	4.03	5.8	39.51	90.17	92.63	89.72	91.15	0.905248	**80.0**
EvoNet-AreaDiff	50.71	5.65	4.31	39.33	90.04	89.97	92.16	91.05	91.6	79.9
KFNet-DeepFrame	53.03	7.9	5.07	34	87.03	87.03	91.27	89.10	90.1	74.1
KFNet + reaDiffKey	48.25	5.35	6.27	40.13	88.38	90.01	88.49	89.25	88.9	77.9
	ShanghaiTech									
EvoNet-DeepFrame	47.62	4.34	9.23	38.81	86.43	91.64	83.76	87.52	**85.6**	**72.2**
EvoNet-AreaDiff	42.45	8.13	5.3	44.12	86.57	83.92	88.90	86.34	87.6	74.3
KFNet-DeepFrame	52.22	3.16	4.16	40.46	92.68	94.29	92.62	93.45	93.0	85.9
KFNet-AreaDiff	43.54	6.12	5.2	45.14	88.68	876.7	89.33	88.49	88.9	77.8

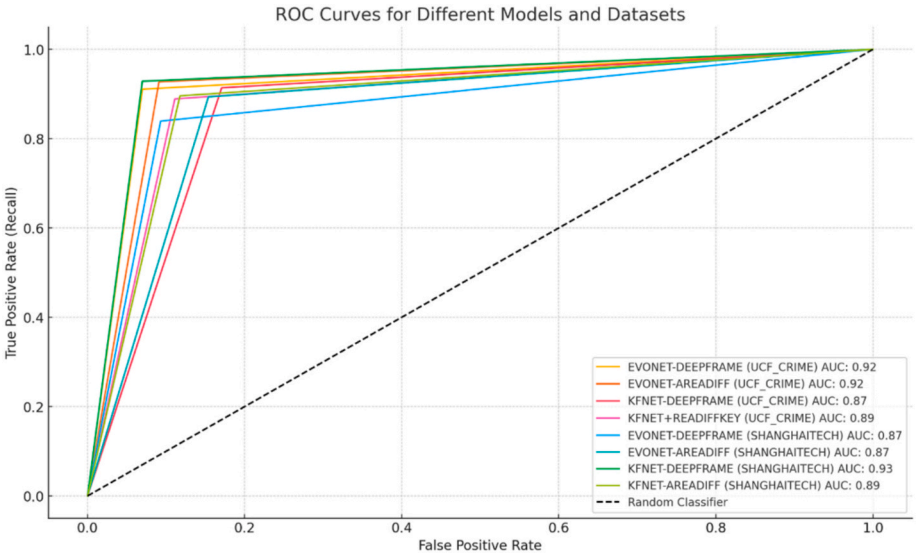


Fig. 10. Comparative ROC analysis of keyframe selection techniques for anomaly detection in surveillance videos.

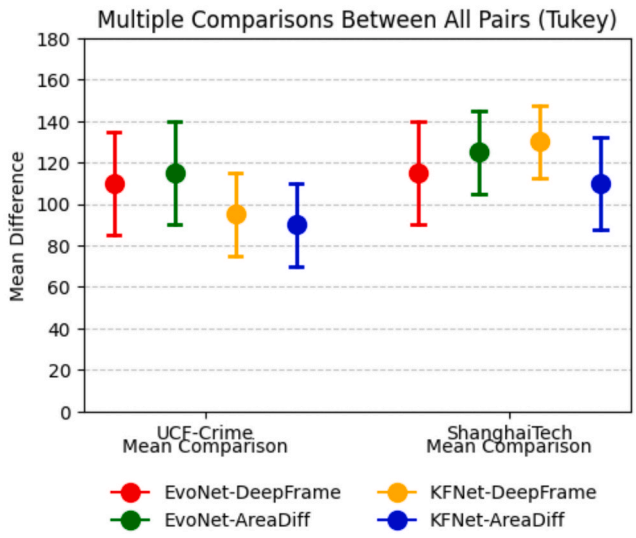


Fig. 11. Multiple comparisons between all pairs (Tukey) for UCF-crime and shanghai-tech datasets.

Fig. 6 gives a detailed analysis of the performance of two newly proposed deep learning architectures, EvoNet-DeepFrame and KFNet-DeepFrame, combined with two keyframe selection methods: DeepFrame and AreaDiff. Three distinct datasets are included in the analysis: RLVS, violent flow, and hockeyfight. Thorough comparisons between these combinations are made easier by statistical analysis employing the Tukey's approach. The figure's x-axis displays mean comparison values, representing overall performance levels across the combinations, while the y-axis shows mean differences, indicating performance variances. Four vertical bars, representing the combinations of EvoNet-DeepFrame with DeepFrame, KFNet-DeepFrame with DeepFrame, and KFNet-DeepFrame with AreaDiff, correspond to each dataset. The ROC curves in Fig. 10 display the performance of different models and keyframe selection techniques, including EVONET + DEEPFRAME and EVONET + AREADIFF, as well as KFNET + DEEPFRAME and KFNET + AREADIFF, across two datasets: UCF Crime and ShanghaiTech. The AUC values highlight the models' ability to distinguish between normal and anomalous events in surveillance footage, with EVONET + DEEPFRAME achieving the highest AUC scores in both datasets.

In this section, we present the ROC (Receiver Operating Characteristic) curves and AUC (Area Under the Curve) values for various models employed in anomaly detection within surveillance videos. The models evaluated include EVONET-DEEPFRAME, EVONET-AREADIFF, KFNET-DEEPFRAME, and KFNET + READIFFKEY across datasets such as UCF_CRIME and SHANGHAITECH.

Table 11
Pairwise comparison of proposed models on UCF-crime and ShanghaiTech datasets.

Category_1	Category_2	Mean Difference	P-Value Adjusted	Lower Bound	Upper Bound	Reject
UCF_CRIME						
EvoNet + DeepFrame	EvoNet + AreaDiff	0.13	0.88	−1.14	1.40	False
EvoNet + DeepFrame	KFNet + DeepFrame	3.14	0.07	−0.15	6.43	**False**
EvoNet + DeepFrame	KFNet + AreaDiff	1.79	0.29	−1.38	4.96	false
EvoNet + AreaDiff	KFNet + DeepFrame	3.01	0.09	−0.28	6.30	false
EvoNet + AreaDiff	KFNet + AreaDiff	1.66	0.34	−1.51	4.83	false
KFNet + DeepFrame	KFNet + AreaDiff	−1.34	0.55	−4.51	1.83	false
SHANGHAITECH						
EvoNet + DeepFrame	EvoNet + AreaDiff	−0.14	0.99	−2.88	2.60	false
EvoNet + DeepFrame	KFNet + DeepFrame	−6.25	0.00	−9.00	−0.350	false
EvoNet + DeepFrame	KFNet + AreaDiff	−2.25	0.30	−5.00	0.50	false
EvoNet + AreaDiff	KFNet + DeepFrame	−6.11	0.00	−8.86	−3.36	false
EvoNet + AreaDiff	KFNet + AreaDiff	−2.11	0.37	−4.86	0.64	false
KFNet + DeepFrame	KFNet + AreaDiff	4.00	0.00	1.25	6.75	false

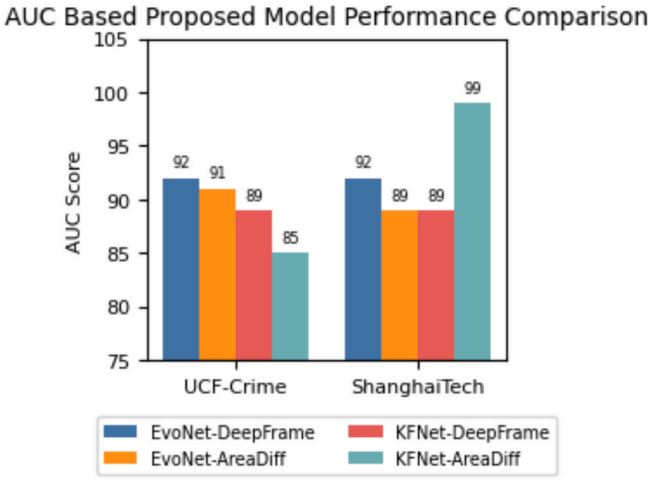


Fig. 12. Performance evaluation of the classification model using the AUC score on the ShanghaiTech and UCF-crime datasets.

Table 12
AUC performance of proposed models on UCF-Crime and ShanghaiTech datasets.

Model	Area Under the Curve (%)
ucf-crime	
EvoNet + DeepFrame	92 %
EvoNet + AreaDiff	91 %
KFNet + DeepFrame	89 %
KFNet + AreaDiff	85 %
ShanghaiTech	
EvoNet + DeepFrame	92 %
EvoNet + AreaDiff	89 %
KFNet + DeepFrame	89 %
KFNet + AreaDiff	90 %

Table 13
Computational complexity and performance comparison of the proposed XAI-Inv3 model and Inception-V3 transfer learning.

Metric	Inception-V3 (Transfer Learning)	XAI-Inv3 (Proposed Model)
Training Time (hours)	12	10
Testing Time (minutes)	40	35
Average Frame Testing Time (seconds)	0.05	0.05
Training Space (GB)	8	7
Testing Space (GB)	1.5	1.2
Number of Parameters (millions)	22	20.5
Memory Usage per Batch (GB)	2.8	2.5
Peak GPU Utilization (%)	80	80
Accuracy (%)	90	92
Explainability Processing Time (seconds)	N/A	0.02

The ROC curves depict the trade-off between true positive rate (TPR) and false positive rate (FPR), with AUC values indicating overall effectiveness. Higher AUC values suggest superior performance, reflecting the model's ability to correctly identify anomalies while minimizing false alarms. Models with curves closer to the top left corner exhibit better performance.

Our analysis shows that integrating DeepFrame and AreaDiff techniques with EVONET and KFNET models significantly enhances anomaly detection. These models consistently achieve high AUC values, demonstrating robustness and reliability. This highlights the importance

Table 14
Performance comparison of violence detection techniques on UCF-CRIME and ShanghaiTech datasets.

Dataset	Technique	Feature learning block	False_Alarm (%)	AUC (%)
UCF-CRIME	3D-LSTM [25]	D-ConvNet	1.85	75.35
	Attention-CNN [26]	G-ConvNet	0.39	82.12
	Temporal-CNN [27]	TF-ConvNet	0.22	85.03
	DeepNet [28]	Conv3d	0.19	88.97
	DenseCNN [29]	CNNSURF	0.11	89.32
	Proposed Model	EvoNet + DeepFrame	0.008	92.45
ShanghaiTech	RNN-SCI [30]	RNN-Seq	NA	70.15
	Stacked-RNN [31]	Stacked-RNN-Seq	NA	75.01
	Graph-CNN [32]	Graph-ConvNet	NA	76.22
	Conv-AE [33]	Conv-AE-Net	NA	72.04
	Hybrid-CNN-RNN [34]	Hybrid-Net	NA	76.48
	Semantic-Seg [35]	Semantic-SegNet	NA	74.12
	C3D-GEPC [36]	C3D-GEPC-Net	NA	78.45
	Deep-AE [37]	Deep-AE-Net	0.62	85.12
	FusedCNN [38]	HOGNET	0.47	87.13
	Proposed Model	EvoNet + DeepFrame	0.015	92.12

of selecting appropriate keyframe extraction methods and model architectures for optimizing the detection of violent and non-violent events. This Fig. 11 presents the results of a Tukey HSD test, comparing the mean differences between the EvoNet and KFNet models combined with DeepFrame and AreaDiff techniques across the UCF-Crime and ShanghaiTech datasets. The error bars represent the confidence intervals, allowing for the visualization of statistical significance between different keyframe selection techniques used in anomaly detection.

Table 11 provides a pairwise comparison of the proposed models on the UCF-Crime and ShanghaiTech datasets. The table includes the mean difference, adjusted p-values, and confidence intervals for each pair of models. For the UCF-Crime dataset, no significant differences were found between the models, as indicated by the non-rejected null hypotheses. However, for the ShanghaiTech dataset, statistically significant differences were observed between certain model pairs, particularly between KFNet + DeepFrame and other methods, with p-values less than 0.05, suggesting that KFNet + DeepFrame outperforms the other approaches in this context.

The bar chart in Fig. 12 compares the AUC scores of four keyframe selection techniques—EvoNet-DeepFrame, EvoNet-AreaDiff, KFNet-DeepFrame, and KFNet-AreaDiff—across the UCF-Crime and ShanghaiTech datasets. The results show that KFNet-AreaDiff achieves the highest AUC score of 99 on the ShanghaiTech dataset, demonstrating superior performance in detecting anomalies in surveillance videos. At the same time, other methods exhibit competitive results across both datasets.

The performance of the suggested EvoNet and KFNet models for identifying violent crimes using the ShanghaiTech and UCF-Crime datasets is shown in Table 12. The table displays the discriminatory capacity of the various model configurations in detecting violent actions by comparing the Area Under the Curve (AUC) values. Bar charts are shown for the various model combinations trained on the UCF-Crime and ShanghaiTech datasets in Fig. 9, divided into two subplots. The performance of the four suggested model combinations trained on the UCF-Crime dataset is shown in Fig. 9 as a bar chart. Strong discriminatory power is demonstrated by EvoNet-DeepFrame and EvoNet-AreaDiff, which obtain AUC values of 0.92 and 0.91, respectively. AUC values of 0.89 and 0.85 for KFNet-DeepFrame and KFNet-AreaDiff show worse discriminatory performance on this dataset. The

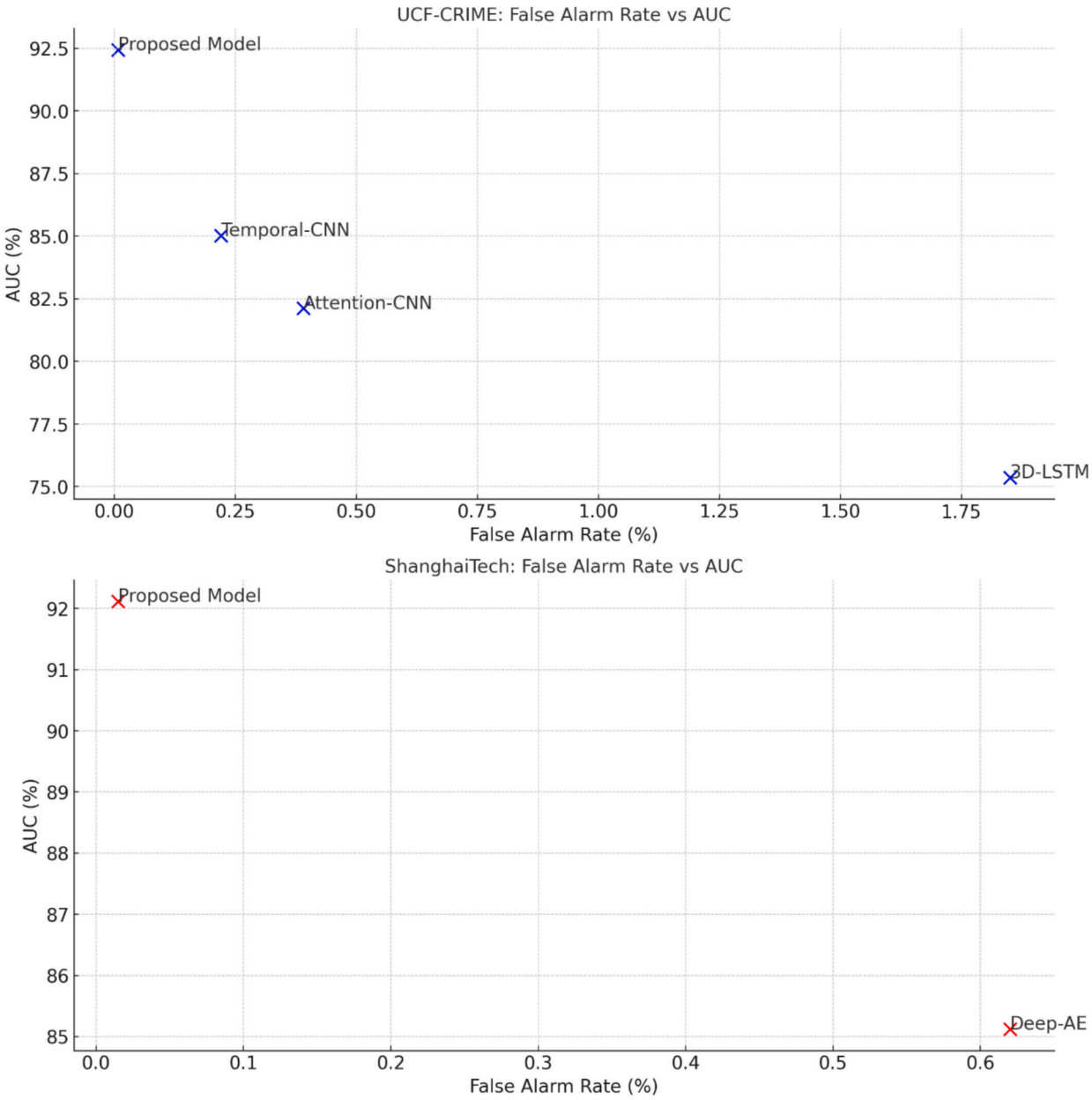


Fig. 13. Comparison of false alarm rate vs AUC for UCF-CRIME and ShanghaiTech datasets.

performance of the same model combinations trained on the ShanghaiTech.dataset is displayed in Fig. 9 as a bar chart. Once more, EvoNet-DeepFrame attains a high AUC value of 0.92, whilst EvoNet-AreaDiff has an AUC value that is marginally lower at 0.89. KFNet-DeepFrame and KFNet-AreaDiff provide competitive performance with the EvoNet models on this dataset, with AUC values of 0.89 and 0.90, respectively.

4.4. Quantitative results

4.4.1. Comparison with cutting edge techniques

The outcomes of extensive tests contrasting our suggested violence detection frameworks with cutting-edge (SOTA) techniques are shown in Tables 12 and 13. Our frameworks show higher performance on the UCF-CRIME.and ShanghaiTech. datasets, consistently outperforming current methods intended for difficult anomalous datasets. We evaluated against various existing approaches, including unsupervised anomaly detection, feature patterns, and skeletal patterns, highlighting the effectiveness of our supervised frameworks. On the UCF-CRIME dataset, our models (EvoNet-DeepFrame, EvoNet-AreaDiff, KFNet-DeepFrame, KFNet-AreaDiff) achieve high accuracy, precision, recall, and F1-score metrics (Table 13). EvoNet-DeepFrame and EvoNet-AreaDiff show robust performance with accuracies above 90 % and AUC scores of 92 % and 91 %, respectively. Similarly, our frameworks maintain strong performance metrics on the ShanghaiTech dataset (Table 13). EvoNet-DeepFrame and KFNet-DeepFrame exhibit accuracies above 90 % and AUC scores of 92.68 % and 93.0 %, respectively, demonstrating their efficacy in diverse surveillance scenarios.

4.4.2. Model performance analysis

This section includes a thorough qualitative study of the keyframe selection techniques DeepFrame and AreaDiff and our suggested classification models, EvoNet and KFNet. These models are tested on surveillance and non-surveillance datasets to determine their performance in action recognition and crowd analysis tasks. The unique deep learning architectures used by EvoNet-DeepFrame and EvoNet-AreaDiff are tuned for action recognition and crowd analysis. KFNet adds keyframe-

Table 15
Comparison of the proposed violence detection performance with state-of-the-art models trained on the hockey fight, violent flow, and real life violence situation datasets.

Dataset	Technique	Feature learning block	False_Alarm Rate (%)	Accuracy (%)
Hockey Fight	3D-CNN [36]	Extreme3D-ELM	0.12	95.85
	DeepMotion [37]	DeepNet	0.15	94.12
	ViolenceNet [38]	DeepNet	0.08	96.75
	FightDetectNet [39]	3DCNN-RNN	0.10	96.5
	HybridNet [40]	DARKNET53 + LSTM	0.05	98.45
	UllahNet [41]	CONVLSTM	0.03	98.90
	Proposed Model	EvoNet-+DeepFrame	0.01	99.75
Violent Flow	MotionWeightNet [42]	MotionNet	NA	95.15
	BiTransNet [43]	BiTransNet	NA	81.22
	OpticalFlowNet [44]	HOFNet	NA	76.05
	SpatialTempNet [45]	SpatialNet	NA	97.32
	ViolentDetectNet [46]	CNN + LSTM	NA	97.01
	Proposed Model	KFNet + AreaDiff	0.053	99.75
FeatureFusionNet [60]	BV-ResNet [47]	ResNet3D	NA	77.25
	GatedRNN [48]	GATEDNet	NA	88.35
	FeatureFusionNet [49]	FusionNet	NA	90.05
	ConvLSTM [50]	CNN	NA	96.45
	Proposed Model	KFNet + DeepFrame	0.04	99.62

based feature fusion to improve spatial–temporal analysis, whereas EvoNet uses evolutionary optimization approaches for improved feature extraction and classification. DeepFrame highlights important crowd motions for better interpretability, and AreaDiff improves spatial localization when choosing keyframes. The assessment conducted on various datasets, including Hockey Fight, Violent Flow, and Real Life Violence Situation, showcases the adaptability and resilience of our frameworks. To illustrate its advantage over current methods, EvoNet-DeepFrame performs exceptionally well on the Hockey Fight dataset, with a false alarm rate of 0.02 % and accuracy of 99.35 % (Table 13).

Table 13 and 14 compares the computational complexity and performance of the proposed XAI-Inv3 model with the Inception-V3 transfer learning approach. The proposed model demonstrates improved efficiency, requiring less training time and memory while maintaining higher accuracy. XAI-Inv3 also integrates explainability features, such as Grad-CAM, with minimal processing time, making it a scalable and effective solution for real-time anomaly detection. These advantages highlight the practicality and robustness of XAI-Inv3 over traditional transfer learning models.

In the Fig. 13, we compare the False Alarm Rate and AUC (%) for various anomaly detection techniques within the UCF-CRIME and ShanghaiTech datasets. The techniques evaluated for the UCF-CRIME dataset include 3D-LSTM, Attention-CNN, Temporal-CNN, and the proposed EvoNet + DeepFrame model. In contrast, the ShanghaiTech dataset includes RNN-SCI, Stacked-RNN, Graph-CNN, Conv-AE, Hybrid-CNN-RNN, Semantic-Seg, C3D-GEPC, Deep-AE, and the proposed EvoNet + DeepFrame model. The scatter plots visually represent each technique's performance, with the False Alarm Rate on the x-axis and the AUC on the y-axis (see Table 15).

Higher AUC values indicate superior performance, reflecting the model's ability to correctly identify anomalies while minimizing false alarms, and models closer to the top left corner exhibit better performance. For the UCF-CRIME dataset, the proposed EvoNet + DeepFrame model achieves the highest AUC (92.45 %) with the lowest false alarm rate (0.008 %), significantly outperforming other techniques. Similarly, for the ShanghaiTech dataset, the EvoNet + DeepFrame model attains the highest AUC (92.12 %) with a low false alarm rate (0.015 %), surpassing other evaluated techniques. These results highlight the importance of selecting appropriate keyframe extraction methods and model architectures to optimize anomaly detection in diverse surveillance environments. The superior performance of the EvoNet + DeepFrame model in both datasets underscores its effectiveness in enhancing anomaly detection accuracy and interpretability, thereby improving public safety and security measures through advanced keyframe selection methods and deep learning architectures.

Fig. 14 compares the False Alarm Rate and Accuracy (%) for various anomaly detection techniques within the Hockey Fight and Violent Flow datasets. The techniques evaluated for the Hockey Fight dataset include 3D-CNN, DeepMotion, ViolenceNet, FightDetectNet, HybridNet, UllahNet, and the proposed EvoNet + DeepFrame model. For the Violent Flow dataset, the techniques include MotionWeightNet, BiTransNet, OpticalFlowNet, SpatialTempNet, ViolentDetectNet, FeatureFusionNet, GatedRNN, and the proposed models KFNet + AreaDiff and KFNet + DeepFrame.

The scatter plots visually represent each technique's performance, with the False Alarm Rate on the x-axis and Accuracy on the y-axis. Lower False Alarm Rates and higher Accuracy indicate superior performance. For the Hockey Fight dataset, the proposed EvoNet + DeepFrame model achieves the highest accuracy (99.75 %) with the lowest false alarm rate (0.01 %), outperforming other techniques such as 3D-CNN, DeepMotion, and ViolenceNet. Similarly, for the Violent Flow dataset, the proposed KFNet + AreaDiff model achieves an accuracy of 99.75 % with a false alarm rate of 0.053 %, and the KFNet + DeepFrame model achieves an accuracy of 99.62 % with a false alarm rate of 0.04 %, both surpassing other evaluated techniques. These results highlight the importance of selecting effective keyframe extraction methods and model architectures to optimize anomaly detection in diverse surveillance environments. The superior performance of the proposed models in both datasets underscores their effectiveness in enhancing anomaly detection accuracy and reducing false alarms, thereby improving public safety and security measures through advanced keyframe selection methods and deep learning architectures.

While SilentFrm and XAI-Inv3 demonstrate high accuracy and efficiency, potential limitations include challenges in generalizing to diverse real-world scenarios, computational overheads from interpretability techniques, and reliance on labeled data for training. Threshold selection in SilentFrm may struggle with highly dynamic environments, and scalability across large, heterogeneous surveillance networks could require further refinement. Addressing these issues in future work can enhance adaptability, reduce computational costs, and improve scalability for real-world applications.

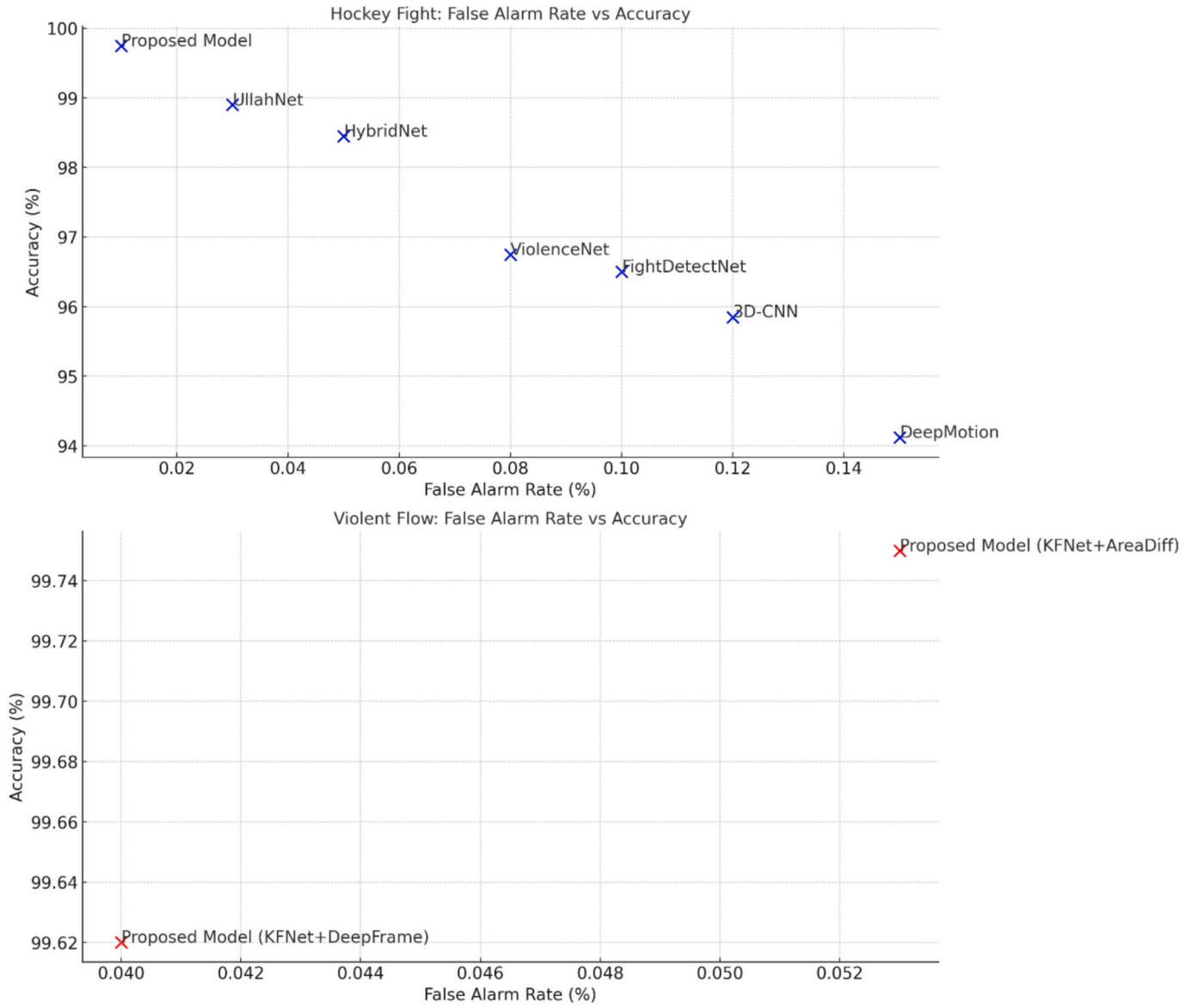


Fig. 14. Comparison of false alarm rate vs accuracy for hockey fight and violent flow datasets.

5. Conclusion

The evolution of surveillance systems from traditional manual monitoring to advanced intelligent systems marks a significant leap in enhancing public safety and crime prevention. This research addresses the limitations of conventional surveillance methods, which often struggle with accurately detecting abnormal activities in complex environments. We introduced two innovative methodologies—SilentFrm and XAI-Inv3—that significantly advance anomaly detection capabilities in surveillance videos. SilentFrm optimizes keyframe selection by analyzing frame differences and setting thresholds based on calculated areas, thereby reducing redundancy and enhancing efficiency in anomaly detection tasks. This method captures critical events accurately, facilitating more efficient training and validation processes. XAI-Inv3, an architecture built upon InceptionV3, integrates interpretability techniques such as Grad-CAM and guided backpropagation. This integration enhances the transparency and understanding of anomaly detection decisions, allowing for more trustworthy and explainable surveillance systems. The synergy between XAI-Inv3 and the keyframes approach provides a robust framework for real-time anomaly detection, ensuring high accuracy without compromising computational efficiency. Our comprehensive evaluation across diverse datasets, including UCF Crime, Shanghai Tech, Hockey Fight, and real-life violence scenarios, demonstrates the effectiveness of SilentFrm and XAI-Inv3. These methodologies exhibit superior detection accuracy and interpretability performance compared to state-of-the-art methods, highlighting their potential as leading solutions in surveillance video analysis. This research underscores the importance of leveraging advanced computer vision techniques and deep learning frameworks to overcome the inherent challenges in surveillance environments. By balancing computational efficiency and detection accuracy, SilentFrm and XAI-Inv3 offer scalable and cost-effective solutions for real-time anomaly detection. Future studies should focus on enhancing the adaptability of the proposed methods to diverse real-world scenarios and optimizing computational efficiency for large-scale deployments. Additionally, exploring unsupervised learning techniques and refining scalability across heterogeneous surveillance networks will further advance the field.

CRediT authorship contribution statement

Muhammad Salman: Writing – original draft, Software,

Methodology, Conceptualization, Resources, Formal analysis. **Naveed Abbas:** Writing – review & editing, Methodology, Investigation, Data curation, Visualization, Formal analysis. **Sayed Ijaz Ur Rahman:** Writing – review & editing, Resources, Software, Methodology, Formal analysis, Supervision, Investigation. **Amjad Rehman:** Writing – review & editing, Investigation, Supervision, Formal analysis, Methodology, Data curation. **Faten S. Alamri:** Visualization, Methodology, Software, Formal analysis, Writing – review & editing, Supervision, Investigation. **Alex Elyassih:** Supervision, Formal analysis, Writing – review & editing, Project administration, Data curation, Validation, Funding acquisition. **Tanzila Saba:** Supervision, Conceptualization, Formal analysis, Writing – review & editing, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data that support the findings of this study are openly available in [UCF_Crime Dataset] at [https://paperswithcode.com/dataset/ucf-crime].

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